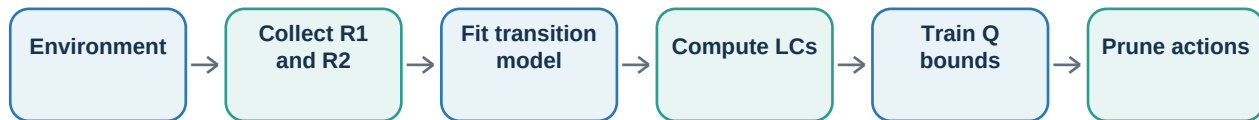


The Smooth-Noise 16-Tile Experiment

A simple guide to the environment, data collection, Wasserstein Lipschitz constants, Q bounds, and action pruning



The one-sentence idea

Every terrain has the same intended movement but a different, smoothly varying noise level. Samples are used to fit that stochastic transition model. Wasserstein distance then compares transition distributions, not unrelated noise draws. Those dynamics constants and reward constants produce learned upper and lower Q bounds; an action is removed only when its best possible value is below another action's guaranteed value.

Scope

This guide describes the code in **dollar_euro_16tile - smooth noise same mean**. It distinguishes the four diagnostic variants from the normal experiment where their estimators differ.

Prepared from the implemented repository setup

September 2026

1. Environment

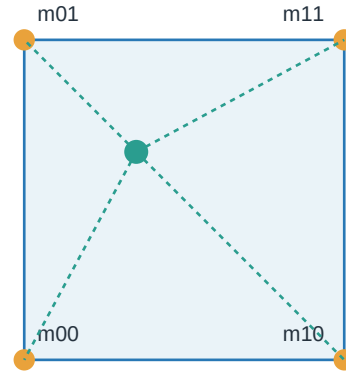
The state is a point $s = (x,y)$ in the unit square $[0,1] \times [0,1]$. The square is divided into 16 equal tiles. Each tile keeps a discrete terrain category for layout, grouping, and plots.

Example at determinism = 0



Dots are tile centers; their sigma targets are interpolation anchors.

One interpolation cell



At (tx, ty) , the four weights are $(1-tx)(1-ty)$, $tx(1-ty)$, $(1-tx)ty$, and $tx ty$.

At the square center: 25% from each anchor.

Actions and rewards

Item	Meaning
Actions	0 Down, 1 Up, 2 Left, 3 Right; intended step length is 0.04.
Start	(0.5, 0.35). Episodes end at Dollar, Euro, Both, or the horizon.
R1	Prefers the Dollar target; also values Both.
R2	Prefers the Euro target; also values Both.
Combined task	Uses $R1 + R2$, so Both is the joint objective.

At determinism = 0, categories 1-4 occupy the four 2×2 quadrants. At determinism > 0, the requested fraction of tiles becomes category 5 and the remaining categories are reproducibly shuffled.

2. Smooth stochastic dynamics

All categories now have the same mean movement. Terrain changes only the standard deviation of isotropic Gaussian noise.

$$s' = \text{clip}(s + 0.04 v[a] + \text{sigma}(s) Z, 0, 1), \quad Z \sim N(0, I_2)$$

Category center	Noise multiplier	Local target standard deviation
1	1.0	1.0 x base sigma
2	1.1	1.1 x base sigma
3	1.2	1.2 x base sigma
4	1.3	1.3 x base sigma
5	deterministic_sigma_scale	scale x base sigma

How interpolation works

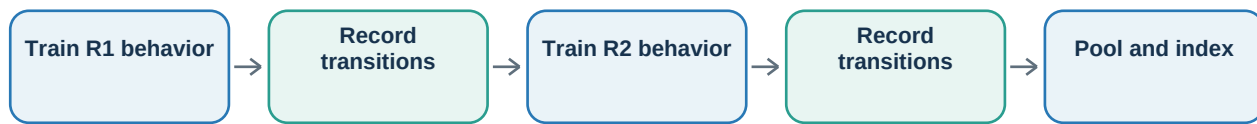
Each tile center is an anchor. Between four neighboring centers, $\text{sigma}(s)$ is their bilinear weighted average. Exactly at a tile center, its assigned value is recovered. Halfway between four centers, each contributes 25%. Outside the outermost centers, the nearest value is extended to the boundary.

Why this change matters

The mean map is a translation, so its Lipschitz constant is 1 away from clipping. The noise scale changes continuously across terrain boundaries, so neighboring transition distributions do not jump abruptly. Direct clipping is non-expansive: it cannot enlarge Euclidean or Wasserstein separation.

Before data collection, the known field gives $K_{\text{sigma}} = \sup \|\text{grad } \text{sigma}(s)\|_2$ and the theoretical global reference $K_P = \sqrt{1 + 2 K_{\text{sigma}}^2}$. The run stops early if $\gamma K_P \geq 1$.

3. What data is collected



- 1. Train behavior R1** - An epsilon-greedy DQN learns using only reward component R1. This makes it visit states relevant to the Dollar preference.
- 2. Record each step** - For every interaction, save source state s , action a , next state s' , reward, source category, source tile, next tile, and whether clipping affected the boundary.
- 3. Train behavior R2** - A second epsilon-greedy DQN uses reward component R2. By default it uses seed + 1,000,000, providing a second behavior distribution.
- 4. Keep free transitions** - Boundary-clipped records are excluded from LC and transition statistics by default. This prevents the wall from being mistaken for unconstrained terrain dynamics.
- 5. Build reusable groups** - The same physical transitions are indexed by source category/action for dynamics, source tile/action for empirical Q checks, and next-state tile for reward slopes. R1 and R2 are stored separately and then combined where required.
- 6. Show coverage** - Four source-state count heatmaps, one per action, combine R1 and R2. Dark cells mean more samples; pale cells reveal sparse coverage.

What --reuse-data means

Reuse occurs only when provenance matches: sigma, gamma, determinism, deterministic scale, steps, seeds, boundary policy, noise-model version, interpolation, multipliers, K_{sigma} , and theoretical K_P . Old hard-boundary transitions are therefore rejected automatically.

4. Fit a stochastic transition model

Raw pairs of sampled successors contain different Gaussian draws. Comparing those draws directly measures sample luck. Instead, the code first estimates the conditional distributions that generated the data.

- 1. Estimate common movement** - For each action a , pool $R1$ and $R2$ across categories and compute $\mu_hat[a] = \text{mean}(s' - s)$. The fitted mean is $f_hat(s,a) = s + \mu_hat[a]$.
- 2. Compute residuals** - For each record, $e = (s' - s) - \mu_hat[a]$. Residuals contain the stochastic part after removing intended movement.
- 3. Fit five center sigmas** - The local sigma is a bilinear mixture of five nonnegative category-center parameters. Maximum likelihood fits these parameters under a 2D isotropic Gaussian model.

For a record i , the fitted local scale σ_i minimizes the Gaussian negative log likelihood: $2 \log(\sigma_i) + \|e_i\|^2 / (2 \sigma_i^2)$.

- 4. Check the fit** - The diagnostic writes fitted action means, fitted center sigmas, normalized residual mean/covariance, a fitted global K_P , and the known theoretical K_P . The theoretical value is a verification reference, not a constant copied into every local row.

Fitted model

$P_hat(. | s,a) = N(s + \mu_hat[a], \sigma_hat(s)^2 I_2)$. The mean displacement is shared; local terrain variation appears through $\sigma_hat(s)$.

If base sigma is zero, the code takes a numerical-floor branch and fits all center sigmas as exactly zero.

5. How the Lipschitz constants are computed

Dynamics LC (Lf or K_P)

For two source states s and t under the same action, compare their fitted Gaussian transition laws with 2-Wasserstein distance:

$$W2^2 = ||s - t||^2 + 2 [\sigma_hat(s) - \sigma_hat(t)]^2$$

$$\text{pair dynamics ratio} = W2 / ||s - t||^2$$

The common mean action displacement cancels. The factor 2 appears because the noise is isotropic in two state dimensions. Clipping is non-expansive, so this pre-clipping comparison is conservative for the clipped kernel.

Reward LC (Lr)

Reward is a function of the next state. For eligible next-state pairs, compute $|R(s1') - R(s2')| / ||s1' - s2'||$. Compute R1 and R2 separately, then add their reward constants for the combined task.

Four diagnostic variants

Variant	Which states are paired	Aggregation
local_mean	Within each category/action; plus neighboring category pairs	Arithmetic mean
local_max	Same local and neighboring groups	Maximum
global_mean	All source states pooled per action	Arithmetic mean
global_max	All source states pooled per action	Maximum

In diagnostics, a pair is eligible only when its denominator distance is at least --min-pair-distance (default 0.04). The gap avoids tiny-denominator amplification; it does not change the fitted stochastic model. In the normal experiment, the corresponding data-fitted ratios use a 10% trimmed mean.

6. Local, cross, and global meaning

Quantity	Grouping	Why it exists
Ordinary Lf	Source category x action	Captures variation among states assigned to one terrain category.
Cross Lf	Neighboring source-category pair x action	Measures smooth transition-distribution change across terrain boundaries.
Global Lf	All source states x action	Tests whether pooling hides or amplifies local differences.
Ordinary Lr	Next-state tile	Reward geometry depends on destination location.
Cross Lr	Physically neighboring next-state tiles	Covers an uncertainty ball that crosses a tile edge.

Combining R1 and R2

For dynamics, the operational combined value is $\max(Lf_R1, Lf_R2)$: the bound must cover either behavior's estimate. For reward, $Lr_total = Lr_R1 + Lr_R2$ because the task reward is their sum.

Bellman O Lipschitz bound

$Lq = Lr Lf / (1 - \gamma Lf)$, valid only when $\gamma Lf < 1$

This follows the discounted Bellman Lipschitz argument: reward sensitivity contributes Lr , transition sensitivity propagates future value by Lf , and repeated discounted propagation forms a geometric series. If $\gamma Lf \geq 1$, the denominator is nonpositive; the code marks that variant invalid and does not train bounds from it.

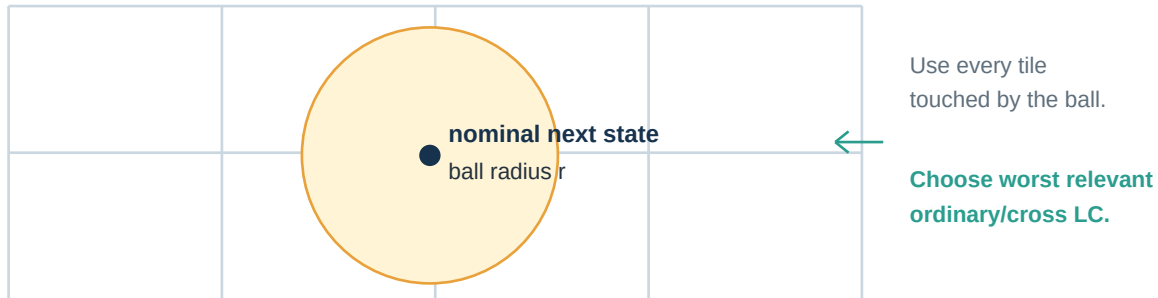
The empirical network slope $|Q(s,a) - Q(t,a)| / ||s - t||$ is saved as a diagnostic. It does not replace the theoretical Bellman Lq used for safe margins.

7. Build a next-state uncertainty ball

For every source category c and current action a , pool free R1 and R2 displacements $\delta = s' - s$.

Stored value	Computation	Purpose
δ_{mean}	pooled mean(δ)	Nominal next-state movement
δ_{std}	pooled sample standard deviation by axis	Observed spread
half-width hw	Student-t multiplier $\times \delta_{\text{std}} \times \sqrt{1 + 1/n}$	95% prediction interval for a new displacement
radius r	$\ hw\ _2$	Circular uncertainty radius used by Q bounds

The nominal state is $s_{\text{bar}}' = \text{clip}(s + \delta_{\text{mean}}[c,a], 0,1)$. The ball $B(s_{\text{bar}}', r)$ describes plausible next states for that current state and action.



Overlap-aware constants

The ball may touch several tiles and categories. The code chooses the largest relevant ordinary or cross-tile L_r . For each possible future action b , it also chooses the largest relevant ordinary or cross-category L_f , then computes $L_{q_eff}(b)$. Finally it takes $\max$ over all four future actions because the Bellman target contains $\max_b Q(s',b)$.

$$\delta_{\text{r}} = L_{\text{r_eff}} r \quad \text{and} \quad \delta_{\text{q}} = \max_b L_{\text{q_eff}}(b) r$$

8. Train upper and lower Q bounds

Two Q networks start from exactly the same weights. One learns an optimistic Bellman target and the other a pessimistic target. Their target networks are updated with Polyak averaging.

$$y_UB = R + \delta_r + \gamma (1-\text{done}) [\max_b Q_UB_target(s_bar',b) + \delta_q]$$

$$y_LB = R - \delta_r + \gamma (1-\text{done}) [\max_b Q_LB_target(s_bar',b) - \delta_q]$$

The reward margin covers moving anywhere inside the next-state ball. The future-value margin covers how much the optimal continuation value can change inside that ball. Larger radius or larger LCs widen the interval; wider intervals lead to less pruning.

Action pruning rule

At state s , first compute $\text{threshold} = \max_b Q_LB(s,b)$. Keep action a when:

$$Q_UB(s,a) \geq \text{threshold} - \text{tolerance}$$

If an action's optimistic value is already below another action's pessimistic value, it cannot be optimal under the learned bounds and is pruned. RA-DQN explores and selects greedily only among surviving actions. During learning, its next-action maximization is also restricted to survivors. A small ranking loss discourages pruned actions from overtaking the best allowed action.

Safety behavior

The LC validity condition is enforced rather than clipped or silently replaced. The action-mask helper also keeps all actions if numerical inconsistency ever produces an empty mask, avoiding an undefined policy decision.

9. End-to-end checklist and outputs

Stage	Input	Main output
1. Environment check	sigma field, gamma	theoretical K_sigma, K_P, gamma K_P
2. R1/R2 collection	two epsilon-greedy behaviors	raw_transitions.npz, coverage heatmaps
3. Transition statistics	free displacement samples	transition_bounds.json
4. Stochastic fit	states and residuals	fitted means/sigmas and verification JSON
5. LC variants	eligible state pairs	ordinary/cross reward and Wasserstein constants
6. Validity	gamma and operational Lf	continue only if gamma Lf < 1
7. Q bounds	LCs and Student-t radii	q_ub_theoretical.pth, q_lb_theoretical.pth
8. Analysis/pruning	frozen Q bounds	value, width, violation, and pruning heatmaps

Diagnostic command

```
run_diagnostics.cmd configs\radial_match.json --sigma 0.0005 --gamma 0.96 --determinism 0.0  
--deterministic-sigma-scale 0.001 --steps 150000 --seed 0 --reuse-data --min-pair-distance 0.04
```

How to read the result

- Coverage first** - Use the shared action heatmaps to identify sparsely sampled areas.
- Fit second** - Check fitted sigmas, normalized residuals, and fitted-vs-theoretical global K_P.
- Compare variants** - Use wasserstein_variant_comparison.json to see whether local/global and mean/max differ materially.
- Check validity** - If gamma Lf ≥ 1 , that variant is skipped. Otherwise inspect bound width and pruning heatmaps.

The key interpretation

The setup separates three uncertainties: stochastic terrain is represented by a fitted transition distribution; finite data creates a Student-t next-state radius; and Lipschitz constants translate that radius into reward and future-value margins. Pruning happens only after all three pieces are combined.